Black Site (Secret Military Facility / Experimental Warfare, Sensors, EW)

Executive Summary

Black Site is a clandestine **military research and operations complex** built for **experimental warfare, advanced sensors, electronic warfare (EW), and autonomous force testing**. The terrain features hardened bunkers, underground labs, sensor arrays, comms vaults, prototype hangars, electromagnetic test chambers, and perimeter kill-zones. The map emphasizes Obsidian Protocol's autonomous EW logic, sensor-fusion under interference, comms-relay disruption, and logistics inside a compartmentalized facility. In AR (Nreal Light/Air), the map anchors at ~4–5 m viewing distance, covering a 10×10 m physical area representing ~1 km² of secret-facility terrain. Key features include EW corridors, stealth-sensor blind zones, prototype weapon platforms, and multi-layer interior combat.

Design Goals & Gameplay

Experimental Warfare & EW Dominance

Black Site prioritizes electronic warfare, sensor disruption, stealth infiltration, and prototype weapon deployment. EW zones dynamically alter unit behavior.

Compartmentalized Interior Combat

The facility is divided into sealed wings, labs, vaults, and hangars. Combat shifts between tight corridors, open testing chambers, and fortified control rooms.

Autonomy & EW-Aware Pathfinding

Autonomous units choose routes based on EW interference, sensor blackout zones, and prototype hazards. Units adapt to shifting electromagnetic conditions.

Sensor & Information Warfare

EMP bursts, jamming fields, thermal distortion, and active countermeasures degrade sensors. Relay towers inside comms vaults restore clarity. Destroying relays creates cascading blackout zones.

Logistics & Resources

Prototype hangars, fuel vaults, and research depots serve as resupply nodes. Convoys must navigate narrow interior lanes and avoid ambush-prone EW corridors.

Deployment & Progress

A fixed deployment budget (~10,000 points) forces players to choose between stealth units, EW specialists, or heavy breach teams. Blue deploys from the southwest access wing; Red deploys from the northeast command vault.

Spatial Constraints (Nreal AR)

Device Capabilities

Nreal Light/Air: ~52° diagonal FOV, 1920×1080 per-eye resolution. Optimal viewing distance ~4–5 m.

Playable Area & Scaling

10×10 m physical area = 1 km² secret-facility zone (1:100 scale). Designed for tabletop AR.

Tracking & Occlusion

Inside-out tracking detects horizontal planes. Depth-mesh (if available) provides limited occlusion. Virtual objects render over real-world view; assume minimal real-world occlusion.

Layout Overview

Coordinate System

1000×1000 grid (1 unit = 1 m). Origin (0,0) = southwest corner.

Key Locations

Blue HQ (Access Wing Command) — (0,0)–150×150 m Security gates, staging area, armored breach vehicles.
Red HQ (Command Vault) — (1000,1000)–150×150 m Hardened vault with EW control nodes and radar mast.
Central Test Chamber — (500,450)–300×300 m Prototype weapons, sensor arrays, open interior combat zone.
Research Lab Cluster — (750,800)–130×130 m Experiment rooms, servers, thermal hazards.
Prototype Hangar — (800,300)–200×200 m Experimental drones, vehicles, interior flanking routes.
EW Corridor — (0,450) → (400,900) Heavy jamming field: extreme sensor degradation.
Maintenance Route — (300,0) → (650,350) Safer interior path with partial cover.
Relay Tower A — (650,150), +240 m Long-range sensor vantage point.
Relay Tower B — (150,650), +360 m Comms relay hub and EW countermeasure node.
Neutral Forward Outpost — (500,950) Drone resupply zone.

Chokepoints & Sightlines

EW Corridor

Severe sensor disruption; ideal for stealth ambushes.

Prototype Hangar

Interior flanking routes; unpredictable hazards.

Test Chamber

Open interior space; long-range experimental weapon lines.

Research Labs

Tight corridors; close-range combat.

Sightlines

Relay Tower A provides long-range sensor coverage (~300 m). EW fields create fog-of-war pockets and blackout zones.

Deployment Zones

Blue Deployment: Qal arrays along a breach routine.
Red Deployment: NE command vault + EW defenses.
Neutral Forward Zone: Outpost at (500,950) for drone drops.

Assets & Environment

Terrain

Labs, vaults, hangars, EW corridors, test chambers, maintenance tunnels.

Vegetation

None — sterile industrial interior.

Buildings & Props

Military: EW towers, radar dishes, AA guns, bunkers. Industrial: Servers, generators, pipelines, control panels. Logistics: Trucks, crates, fuel drums, drones.

Effects

EMP bursts, sparks, smoke, heat shimmer, holographic displays.

Audio

Machinery hum, alarms, radio chatter, ventilation systems.

LOD & Budget

Asset Cate	Count	Tri/Item	Total Tris
Heavy Mo	3	200k	600k
Industrial	4	80k	320k
Buildings/	15	50k	750k
Corridors	4	30k	120k
Props	50	10k	500k
Total	—	—	**~2.29M**

Texture Budget

2048² for terrain/buildings, 1024² for props. DXT1/5 compression. Target ≤100 MB.

Unreal / Nreal Integration

Engine & Plugins

Unreal ARTemplate + Nreal SDK (XREAL). ARKit/ARCore enabled.

Rendering

Forward renderer, stationary lights, baked GI, aggressive culling.

Anchors

Single spatial anchor for entire map.

Interaction

Phone pointer for pan/zoom; HUD as screen-space widgets.

Multiplayer

Client-server architecture; offload AI/physics to server.

Implementation Plan & Testing

Milestones

1. **Prototype Layout — Month 1** Block-out labs, hangars, EW corridor. AR anchor test.
2. **Core Systems — Month 2** NavMesh. EW-aware autonomy behaviors. sensor logic.
3. **AR Interaction — Month 3** Anchors, plane detection, basic interactivity.
4. **Content Fill — Month 4–5** Final models, textures, lighting, audio.
5. **Gameplay & Polish — Month 6–7** Objectives, balance, VFX/EMP bursts, sparks, holograms).
6. **Testing & QA — Month 8** Multiplayer stress test, sensor validation.

Testing Plan

Autonomy

Verify EW-aware navigation, blackout-zone behavior, hazard avoidance.

Comms/Sensors

Test relay destruction, fog-of-war updates, LOS checks under EW.

Performance

≥60 Hz AR render on target device.

QA Checklist

Navigation, EW logic, deployment budget enforcement, anchor drift, sensor accuracy.